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SOLOR TRACKER AND CONTROL METHOD OF THE SOLAR TRACKER

Abstract

A solar tracker that includes a first and second sets of panels, a first motor coupled to the first set of panels for moving the same to a final angular position, and a second motor coupled to the second set of panels for moving the same to the final angular position. A first sensor determines an initial angular position of the first set of panels, and a second sensor determines an initial angular position of the second set of panels. A first controller is used to control the first motor to move the first set of panels to the final angular position depending on the initial angular position determined by the first sensor. A second controller is used to control the second motor to move the second set of panels to the final angular position depending on the initial angular position determined by the second sensor.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application relates to and claims the benefit and priority to International Application. No. PCT/ES2023/070720, filed Dec. 1, 2023, which claims the benefit and priority to European Application No. EP22383247.8, filed Dec. 21, 2022.

FIELD
[0002] The present invention relates to solar trackers and control methods of the operation of said trackers.
BACKGROUND
[0003] Solar trackers orient photovoltaic solar panels towards the sun in order to track the sun's trajectory throughout the day and maximize energy capture. Generally, the efficiency of photovoltaic solar panels is maximized when they receive solar radiation perpendicular to the surface of the panels.
[0004] Horizontal solar trackers rotate the photovoltaic solar panels about an axis of rotation which is substantially horizontal with respect to the ground. The axis of rotation is aligned in the north-south direction and rotates in the east-west direction for solar radiation to strike as perpendicular as possible to the surface of the photovoltaic solar panels throughout the day. In high-capacity solar plants, horizontal solar trackers have several axes and are arranged in several rows of photovoltaic solar panels, with one axis per row.
[0005] Horizontal solar trackers comprise a plurality of posts extending and separated longitudinally along an axis and a support structure which is supported on the plurality of posts. The support structure comprises a main crossbar extending along the axis and a plurality of transverse crossbars oriented orthogonally to the main crossbar. A plurality of photovoltaic solar panels is arranged on the support structure and a rotation is applied to the main crossbar to move the photovoltaic solar panels and orient same towards the sun by means of an electric motor which is operatively coupled to the main crossbar.
[0006] There are different mechanical solutions to cause the rotational movement of the main crossbar, seeking to maximize the total number of photovoltaic solar panels that are moved with a single electric motor at all times. This is usually achieved with complex mechanical systems which allow movement to be transmitted from a single motor. For example, WO2000063625A1 shows a solar tracker with several rows of photovoltaic solar panels which uses transverse coupling rods to transmit the movement of an electric motor to all the main crossbars of the solar tracker. WO2019183492A1 shows a solar tracker with several linear actuators distributed along the axis of the solar tracker to apply a rotation on the main crossbar, and the movement of an electric motor is transmitted to the linear actuators by means of longitudinal coupling systems.
[0007] US20200304059A1 shows a solar tracker comprising a plurality of photovoltaic solar panels with a first set of panels and a second set of panels, a plurality of posts extending and separated longitudinally along an axis, a support structure which is supported on the plurality of posts, the support structure comprises a main crossbar extending along the axis and a plurality of transverse crossbars oriented orthogonally to the main crossbar, the plurality of photovoltaic solar panels is arranged on the support structure, a first electric motor which is operatively coupled to a

first part of the main crossbar to apply a rotation to the first part of the main crossbar and move the first set of panels to a final angular position, and a second electric motor which is operatively coupled to a second part of the main crossbar, spaced from the first part, to apply a rotation to the second part of the main crossbar and move the second set of panels to the final angular position. In that sense, document US20200304059A1 shows a solar tracker with several electric motors distributed along the main crossbar of the solar tracker. The electric motors are controlled by means of a single central controller and attached by means of a power line which applies a voltage to all the electric motors at the same time so that they can act simultaneously.

SUMMARY

[0008] Disclosed is a solar tracker and a control method of the solar tracker.

[0009] An aspect of the invention relates to a solar tracker comprising a plurality of photovoltaic solar panels with a first set of panels and a second set of panels; a plurality of posts extending and separated longitudinally along an axis; a support structure which is supported on the plurality of posts, the support structure comprises a main crossbar extending along the axis and a plurality of transverse crossbars oriented orthogonally to the main crossbar, the plurality of photovoltaic solar panels is arranged on the support structure; a first electric motor which is operatively coupled to a first part of the main crossbar to apply a rotation to the first part of the main crossbar and move the first set of panels to a final angular position; a second electric motor which is operatively coupled to a second part of the main crossbar, spaced from the first part, to apply a rotation to the second part of the main crossbar and move the second set of panels to the final angular position; a first position sensor for determining an initial angular position of the first set of panels; a first controller configured to control the first electric motor and move the first set of panels to the final angular position depending on the initial angular position determined by the first position sensor; a second position sensor for determining an initial angular position of the second set of panels; and a second controller configured to control the second electric motor and move the second set of panels to the final angular position depending on the initial angular position determined by the second position sensor.

[0010] Another aspect of the invention relates to a control method of the solar tracker defined above, which comprises: [0011] selecting the final angular position of the plurality of photovoltaic solar panels, [0012] determining the initial angular position of the first set of panels, [0013] determining the initial angular position of the second set of panels, [0014] moving the first set of panels to the final angular position depending on the initial angular position of the first set of panels, and [0015] moving the second set of panels to the final angular position depending on the initial angular position of the second set of panels.

[0016] A solar tracker having a controller for each electric motor, which allows each electric motor to act independently on its respective set of panels, is thereby obtained. Furthermore, each controller has information about the angular position to which the set of panels must be moved, whereby the angular position of each set of panels can be corrected in the event that the sets of panels are misaligned with respect to one another. This independent control of the panels cannot be achieved in systems using a single motor which move all the photovoltaic solar panels at the same time by means of mechanical systems (WO2000063625A1, WO2019183492A1), or in solar trackers using a single controller to move several electric motors which act on the photovoltaic panels at the same time (US20200304059A1).

[0017] Furthermore, having information about the angular position of each set of panels separately allows the galloping effect to be detected. Said effect is an aeroelastic instability caused by wind which generates large-amplitude oscillations in the support structures of solar trackers, causing a movement that may end with the structure collapsing. By knowing the angular position of each set of panels and being able to move said panels independently, the sets of panels can be moved to angular positions in which said effect is reduced or cancelled.

[0018] These and other advantages and features will become apparent in view of the figures and detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0019] FIGS. **1** and **2** show schematic views of an embodiment of a solar tracker with a row of photovoltaic solar panels and two sets of panels in the row.
[0020] FIG. **3** shows a schematic view of an embodiment of a solar tracker with three rows of photovoltaic solar panels and more than two sets of panels in each row.
[0021] FIG. **4** shows a diagram of controllers according to one embodiment that are used to control the angular position of the photovoltaic solar panel.
[0022] FIG. **5** shows a diagram of controllers according to another embodiment that are used to control the angular position of the photovoltaic solar panel.
DETAILED DESCRIPTION
[0023] The invention relates to a solar tracker comprising a plurality of photovoltaic solar panels **10**.
[0024] The plurality of photovoltaic solar panels **10** has at least a first set of panels **11** and a second set of panels **12**. A first electric motor **13** is operatively coupled to the first set of panels **11** to move the first set of panels **11** to a final angular position and a second electric motor **14** is operatively coupled to the second set of panels **12** to move the second set of panels **12** to the final angular position.
[0025] The angular position of a set of panels is the angle defined between the horizontal of the ground and the surface of the photovoltaic solar panels **10** of the set of panels.
[0026] The final angular position may be the angular position of the photovoltaic solar panels **10** in which the conversion of solar radiation into electric power is maximized. For example, the final angular position is a position in which the surface of the photovoltaic solar panels **10** is comprised in a plane substantially perpendicular to solar radiation.
[0027] The solar tracker comprises a first position sensor **15** for determining an initial angular position of the first set of panels **11** and a second position sensor **16** for determining an initial angular position of the second set of panels **12**. The solar tracker also comprises a first controller **17** which is configured to control the first electric motor **13** and move the first set of panels **11** to the final angular position depending on the initial angular position determined by the first position sensor **15**, and a second controller **18** which is configured to control the second electric motor **14** and move the second set of panels **12** to the final angular position depending on the initial angular position determined by the second position sensor **16**.
[0028] In this way, each controller **17** and **18** knows the actual angular position of its respective set of panels **11** and **12** at all times and may calculate the angle at which each set of panels **11** and **12** must be moved in order to reach the final angular position. When the controller determines the angle, the controller calculates the speed at which the set of panels must be moved and powers the motor to move the panels at the calculated speed until reaching the required angular position. The operation of each controller **17** and **18** is described below.
[0029] Preferably, the first controller **17** is a master controller and the second controller **18** is a slave controller, the first controller **17** and the second controller **18** being connected to one another through a data and power supply line **35**. In this way, although each controller controls its respective electric motor independently, the master controller **17** may monitor all the operations of the slave controller **18**, and the master controller **17** may send orders to the slave controller **18** and receive data from the slave controller **18**.
[0030] The solar tracker comprises a plurality of posts **19** extending and separated longitudinally

along at least one axis X and at least one support structure **20** which is supported on the plurality of posts **19**. Each support structure **20** comprises a main crossbar **21** extending along the axis X and a plurality of transverse crossbars **22** oriented orthogonally to the main crossbar **21**.

[0031] Preferably, the position sensors **15** and **16** are accelerometers. Accelerometers measure linear accelerations in one axis, two axes, or three axes.

[0032] The first accelerometer **15** is arranged in a part of the support structure **20** that supports the first set of panels **11** and the second accelerometer **16** is arranged in a part of the support structure **20** that supports the second set of panels **12**. This ensures that the accelerometers are arranged on the support structure **20** of the photovoltaic solar panels **10**. This favors the detection of the galloping effect since measurement is performed directly in the place where the effect occurs.

[0033] Even more preferably, the first accelerometer **15** is arranged in an enclosing casing of the first controller **17** which is in turn arranged in a part of the support structure **20** that supports the first set of panels **11**, and the second accelerometer **16** is arranged in an enclosing casing of the second controller **18** which is in turn arranged in a part of the support structure **20** that supports the second set of panels **12**. This allows the casing of the controller to hold the accelerometer, whereby a compact device which houses the electronics for controlling the electric motor is obtained.

[0034] The schematic example of FIGS. **1** and **2** shows a solar tracker with a row of photovoltaic solar panels **10**. The solar tracker comprises a plurality of posts **19** extending and separated longitudinally along an axis X and a support structure **20** which is supported on the plurality of posts **19**. The support structure **20** comprises a main crossbar **21** extending along the axis X and a plurality of transverse crossbars **22** oriented orthogonally to the main crossbar **21**. The plurality of photovoltaic solar panels **10** is arranged on the support structure **20**, the first electric motor **13** is operatively coupled to a first part **23** of the main crossbar **21** to apply a rotation to the first part **23** of the main crossbar **21** and move the first set of panels **11** to the final angular position, and the second electric motor **14** is operatively coupled to a second part **24** of the main crossbar **21**, spaced from the first part **23**, to apply a rotation to the second part **24** of the main crossbar **21** and move the second set of panels **12** to the final angular position.

[0035] Preferably, the first electric motor **13** is operatively coupled to a first actuator **25** which is arranged on the first part **23** of the crossbar longitudinal **21** and the second electric motor **14** is operatively coupled to a second actuator **26** which is arranged on the second part **24** of the crossbar longitudinal **21**. The actuator **25** or **26** may be a slew drive, a linear actuator, or any type of actuator commonly used in the photovoltaic sector for coupling to the main crossbar **21** of the support structure **20** of a solar tracker and rotating the main crossbar **21**.

[0036] The solar tracker may comprise more than two sets of panels, and each set of panels has an electric motor for moving the set of panels, a position sensor for determining the initial angular position of the set of panels, and a controller for controlling the electric motor and moving the set of panels to the final angular position depending on the initial angular position determined by the position sensor. Furthermore, each motor is operatively coupled to an actuator. For example, FIG. **3** shows a solar tracker with three rows of photovoltaic solar panels **10** and each row has three sets of solar panels. The solar tracker has a first controller **17** and several second controllers **18** and **18′**, wherein the first controller **17** is a master controller of the solar tracker and the second controllers **18** and **18′** are slave controllers. The master controller **17** is connected to each of the slave controllers **18** by means of a respective data and power supply line (which for the sake of clarity is not depicted in FIG. **3**). Preferably, the master controller **17** is located in the center of the solar tracker.

[0037] The plurality of photovoltaic solar panels **10** is connected to a power line **27** to extract the electric power generated by the photovoltaic solar panels **10**, and the first electric motor **13** and the second electric motor **14** are connected to the power line **27** to receive electric power from the power line **27**. In this way, the electric power generated by the photovoltaic solar panels **10** is utilized to move the sets of panels **11** and **12**. The power line **27** is commonly known in the photovoltaic sector as the solar tracker string, and it is the line through which the electric power generated by each photovoltaic solar panel **10** is extracted, which power is generally delivered to a string box, from which electricity is delivered to the grid.

[0038] FIGS. **4** and **5** show two examples for connecting the controllers **17** and **18** to the power line **27** and controlling the electric motors **13** and **14**.

[0039] As shown in FIG. **4**, the power line **27** extracting the electric power generated by the photovoltaic solar panels **10** is the data and power supply line **35** which is used for transmitting electric power and communication data between the first controller **17** and the second controller **18**. In other words, the same physical means are utilized to transmit electric power and communication data between the controllers. In other words, the solar tracker string is used to connect the controllers **17** and **18** to one another, thereby reducing the tracker cost, since the string is a wiring that is always required in the photovoltaic solar panels **10** for power extraction.

[0040] As shown in FIG. **5**, the first controller **17** is connected to the power line **27** of the plurality of photovoltaic solar panels **10** to receive electric power from the power line **27**, and the second controller **18** is connected to the first controller **17** to receive electric power and communication data from the first controller **17** through the data and power supply line **35**. In this case, unlike FIG. **4**, a dedicated umbilical cable **35** is used to transmit electric power and communication data between the controllers **17** and **18**. This option of FIG. **5** uses more wiring than the option of FIG. **4**, but allows the second controller **18** to have simpler electronics that are therefore more cost-effective, since it does not require adapting the power from line **27**, given that said function is performed by the first controller **17**. For example, line **27** has a voltage of 1500 V, and the first controller **17** transforms the voltage from 1500 V to 24 V.

[0041] As seen in FIGS. **4** and **5**, the first controller **17**, acting as the master controller, comprises a DC/DC converter **28**, a power unit PWR **29**, a microcontroller **30**, a driver **31**, and the first accelerometer **15**. Furthermore, the first controller **17** has a battery BAT **32**, a communication interface HMI **33**, and a communications port COM **34**.

[0042] The DC/DC converter **28** adapts the electric power from the power line **27** so that it may be used by the different components of the first controller **17**. The power unit PWR **29** receives the electric power from the DC/DC converter **28** and adapts it so that it can be used by the microcontroller **30**.

[0043] The microcontroller **30** of the first controller **17** calculates the final angular position to which all the sets of panels **11** and **12** of the solar tracker must be moved. The microcontroller **30** receives the initial angular position measured by the first accelerometer **15** and sends a setpoint signal to the driver **31** of the first electric motor **13** for it to act on the first actuator **25** and move the first set of panels **11** to the final angular position. The setpoint signal sent to the driver **31** comprises a position signal, a speed signal, and a direction of rotation, and based on said signals, the driver **31** powers the first electric motor **13**. The position signal has the final angular position to which the first set of panels **11** must be moved, and the speed signal has the speed at which the first set of panels **11** must be moved in order to reach the final angular position.

[0044] The final angular position may be calculated in real time by the microcontroller **30** of the first controller **17** based on an astronomical algorithm. In other words, the microcontroller **30** may calculate the position of the sun depending on the location of the solar tracker (latitude and longitude) and the time of the day. The final angular position may also be sent to the microcontroller **30** by an operator from the communication interface HMI **33**, for example, to perform maintenance tasks when the photovoltaic solar panels **10** are to be moved to a specific position.

[0045] The battery BAT **32** is optional and may be used to send electric power to the motor in the event that the power obtained from the power line **27** is not enough, which occurs, for example, at night when the panels **10** do not produce any power. The communications port **34** comprises high-level communication protocols such as, for example, ZigBee or Bluetooth, used by the master controller **17** to generate the high-level communication network. The controller **17** may further have a service communication port for being connected with the microcontroller **30** in the event that the communications port **34** fails.

[0046] The second controller **18**, acting as the slave controller, comprises the same elements as the first controller **17**, acting as the master, with the exception that it has no communication interface HMI **33** nor communications port COM **34**.

[0047] The microcontroller **30** of the second controller **18** receives the final angular position to which the second set of panels **12** must be moved from the microcontroller **30** of the first controller **17**. The microcontroller **30** of the second controller **18** also receives the initial angular position measured by the second accelerometer **16**. The microcontroller **30** of the second controller **18** also receives the speed at which the second set of panels **12** must be moved in order to reach the final angular position from the microcontroller **30** of the first controller **17**. Based on the foregoing, the microcontroller **30** of the second controller **18** sends a setpoint signal to the driver **31** of the second electric motor **14** for it to act on the second actuator **26** and move the second set of panels **12** to the final angular position. The setpoint signal sent to the driver **31** of the second controller **18** also comprises a position signal, a speed signal, and a direction of rotation, and based on said signals, the driver **31** powers the second electric motor **14**. The position signal has the final angular position to which the second set of panels **12** must be moved, and the speed signal has the speed at which the second set of panels **12** must be moved in order to reach the final angular position.

[0048] As also shown in FIG. **5**, the second controller **18** also has no DC/DC converter **28** nor a battery **28**, since in FIG. **5**, the power unit PWR **29** of the second controller **18** receives the electric power from the DC/DC converter **28** of the first controller **17** through the data and power supply line **35**. This allows the electronics of the second controller to be simpler and therefore more cost-effective.

[0049] In this way, the control method of the solar tracker comprises the following steps: [0050] selecting the final angular position of the plurality of photovoltaic solar panels **10**, [0051] determining the initial angular position of the first set of panels **11** by means of the first position sensor **15**, [0052] determining the initial angular position of the second set of panels **12** by means of the second position sensor **16**, [0053] moving the first set of panels **11** to the final angular position depending on the initial angular position of the first set of panels **11**, and [0054] moving the second set of panels **12** to the final angular position depending on the initial angular position of the second set of panels **12**.

[0055] To move the first set of panels **11** to the final angular position, the first controller **17** receives the initial angular position of the first set of panels **11** measured by the position sensor **15** and determines the angle at which the first set of panels **11** must be moved, for example, by subtracting the initial angular position from the final angular position.

[0056] Then, the first controller **17** determines the direction of rotation and the speed at which the first set of panels **11** must be moved in order to reach the final angular position and powers the first electric motor **13** in order to reach said final angular position.

[0057] To move the second set of panels **12** to the final angular position, the second controller **18** receives the final angular position from the first controller **17**, receives the initial angular position of the second set of panels **12** measured by the second position sensor **16**, and determines the angle at which the second set of panels **12** must be moved, for example, by subtracting the initial angular position from the final angular position.

[0058] The second controller **18** also receives the speed at which the second set of panels **12** must be moved in order to reach the final angular position from the first controller **17** and powers the second electric motor **14** in order to reach said final angular position. The direction of rotation may be sent to the second controller **18** from the first controller **17**, or the second controller **18** itself may determine the direction of rotation.

[0059] The first controller **17**, acting as the master, monitors the operation of the second controller

18, acting as the slave, in real time. To that end, the second controller **18** sends data about the angular position of the second set of panels **12** to the first controller **17**. The second controller **18** also sends data about the speed at which the second set of panels **12** is moved to the first controller **17**.

[0060] For example, if the two sets of panels **11** and **12** were to be moved to a final angular position of 45°, with both sets **11** and **12** being in an initial angular position of 40°, the controllers **17** and **18** may send a single order to the electric motors **13** and **14** for them to move the sets of panels **11** and **12** from 40° to 45°.

[0061] When the angle between the initial angular position and the final angular position is less than a certain angle, for example less than 5°, the photovoltaic solar panels **10** may be moved between the initial angular position and the final angular position in a single movement, however, when the angle between the initial angular position and the final angular position is greater than a certain angle, for example greater than 5°, the photovoltaic solar panels **10** are preferably moved between the initial angular position and the final angular position in more than one movement, being established to that end partial angular positions.

[0062] According to the foregoing, after selecting the final angular position of the plurality of photovoltaic solar panels **10**, the method comprises the following steps: [0063] determining the initial angular position of the first set of panels **11**, [0064] determining the initial angular position of the second set of panels **12**, [0065] selecting a partial angular position of the plurality of photovoltaic solar panels **10**, the partial angular position being comprised between at least one of the initial angular positions and the final angular position, [0066] moving the first set of panels **11** to the partial angular position depending on the initial angular position of the first set of panels **11**, [0067] moving the second set of panels **12** to the partial angular position depending on the initial angular position of the second set of panels **12**, and [0068] repeating the preceding steps until reaching the final angular position of the plurality of photovoltaic solar panels **10**.

[0069] For example, if the sets of panels **11** and **12** were to be moved to a final angular position of 45°, with both sets of panels **11** and **12** being in an initial angular position of 0°, the controllers **17** and **18** may send partial orders to the motors **13** and **14** for them to move the sets of panels **11** and **12** from 0° to 5°, then from 5° to 10°, then from 10° to 15°, and repeating the process until reaching 45°. This option allows minimizing the risk of misalignments between the sets of panels **10** and **11**.

[0070] According to one embodiment, the first set of panels **11** and the second set of panels **12** are moved simultaneously until reaching the final angular position, i.e., all the photovoltaic solar panels **10** are moved at the same time.

[0071] According to another embodiment, the first set of panels **11** and the second set of panels **12** are moved sequentially until reaching the final angular position, wherein first the photovoltaic solar panels **10** of one set of panels **12** are moved, and then the photovoltaic solar panels **10** of the other set of panels **11** are moved.

[0072] Preferably, when the photovoltaic solar panels **10** of the two sets of panels **11** and **12** are arranged on the same main crossbar **21** of the support structure **20**, the control method comprises the following steps: [0073] determining a maximum torsion allowed by the main crossbar **21** of the support structure **20** on which the plurality of photovoltaic solar panels **10** is arranged, [0074] applying a first rotation to the first part **23** of the main crossbar **21** to move the first set of panels **11** without the first rotation exceeding the maximum torsion allowed by the main crossbar **21**, [0075] applying a second rotation to the second part **24** of the main crossbar **21** to move the second set of panels **11** without the second rotation exceeding the maximum torsion allowed by the main crossbar **21**, and [0076] repeating the application of the first rotation and the second rotation until reaching the final angular position of the plurality of photovoltaic solar panels **10**.

[0077] In this way, instead of moving the actuators **25** and **26** at the same time, the actuators **25** and **26**, and therefore the sets of panels **11** and **12**, are moved sequentially, at intervals, which allows the maximum electric power required for moving the photovoltaic solar panels **10** of the solar tracker to be reduced, in exchange for a small torsion in the main crossbar **21** of the support structure **20**, and generally a lower advancement speed.

[0078] The maximum torsion allowed by the main crossbar **21** is determined according to the mechanical properties of the main crossbar **21**. Generally, the crossbar **21** is a tubular profile with a square section, so the type of material of the tubular profile and the thickness of the profile walls must be taken into account, among other aspects, in order to determine the maximum torsion.

[0079] Embodiments are disclosed in the clauses that follow.

[0080] Clause 1. A solar tracker comprising: [0081] a plurality of photovoltaic solar panels (**10**) with a first set of panels (**11**) and a second set of panels (**12**); [0082] a plurality of posts (**19**) extending and separated longitudinally along an axis (X); [0083] a support structure (**20**) which is supported on the plurality of posts (**19**), the support structure (**20**) comprises a main crossbar (**21**) extending along the axis (X) and a plurality of transverse crossbars (**22**) oriented orthogonally to the main crossbar (**21**), the plurality of photovoltaic solar panels (**10**) is arranged on the support structure (**20**); [0084] a first electric motor (**13**) which is operatively coupled to a first part (**23**) of the main crossbar (**21**) to apply a rotation to the first part (**23**) of the main crossbar (**21**) and move the first set of panels (**11**) to a final angular position; and a second electric motor (**14**) which is operatively coupled to a second part (**24**) of the main crossbar (**21**), spaced from the first part (**23**), to apply a rotation to the second part (**24**) of the main crossbar (**21**) and move the second set of panels (**12**) to the final angular position; [0085] a first position sensor (**15**) for determining an initial angular position of the first set of panels (**11**); [0086] a first controller (**17**) configured to control the first electric motor (**13**) and move the first set of panels (**11**) to the final angular position depending on the initial angular position determined by the first position sensor (**15**); [0087] a second position sensor (**16**) for determining an initial angular position of the second set of panels (**12**); and [0088] a second controller (**18**) configured to control the second electric motor (**14**) and move the second set of panels (**12**) to the final angular position depending on the initial angular position determined by the second position sensor (**16**).

[0089] Clause 2. The solar tracker according to clause 1, wherein the plurality of photovoltaic solar panels (**10**) is connected to a power line (**27**) to extract the electric power generated by the photovoltaic solar panels (**10**), and the first electric motor (**13**) and the second electric motor (**14**) are connected to the power line (**27**) to receive electric power from the power line (**27**).

[0090] Clause 3. The solar tracker according to clause 1 or 2, wherein the first controller (**17**) is a master controller and the second controller (**18**) is a slave controller, the first controller (**17**) and the second controller (**18**) being connected to one another through a data and power supply line (**35**).

[0091] Clause 4. The solar tracker according to the preceding clause, wherein the power line (**27**) extracting the electric power generated by the photovoltaic solar panels (**10**) is the data and power supply line (**35**) which is used for transmitting electric power and communication data between the first controller (**17**) and the second controller (**18**).

[0092] Clause 5. The solar tracker according to clause 3, wherein the first controller (**17**) is connected to the power line (**27**) of the plurality of photovoltaic solar panels (**10**) to receive electric power from the power line (**27**), and the second controller (**18**) is connected to the first controller (**17**) to receive electric power and communication data from the first controller (**17**) through the data and power supply line (**35**).

[0093] Clause 6. The solar tracker according to any of the preceding clauses, wherein the position sensors (**15**, **16**) are accelerometers.

[0094] Clause 7. The solar tracker according to the preceding clause, wherein the first accelerometer (**15**) is arranged in a part of the support structure (**20**) that supports the first set of panels (**11**), and the second accelerometer (**16**) is arranged in a part of the support structure (**20**) that supports the second set of panels (**12**).

[0095] Clause 8. The solar tracker according to the preceding clause, wherein the first accelerometer (**15**) is arranged in an enclosing casing of the first controller (**17**) which is in turn

arranged in a part of the support structure (**20**) that supports the first set of panels (**11**), and the second accelerometer (**16**) is arranged in an enclosing casing of the second controller (**18**) which is in turn arranged in a part of the support structure (**20**) that supports the second set of panels (**12**).
[0096] Clause 9. A control method of the solar tracker according to any of the preceding clauses, comprising the following steps: [0097] selecting the final angular position of the plurality of photovoltaic solar panels (**10**); [0098] determining the initial angular position of the first set of panels (**11**); [0099] determining the initial angular position of the second set of panels (**12**); [0100] moving the first set of panels (**11**) to the final angular position depending on the initial angular position of the first set of panels (**11**); and [0101] moving the second set of panels (**12**) to the final angular position depending on the initial angular position of the second set of panels (**12**).
[0102] Clause 10. The control method according to the preceding clause, wherein, after selecting the final angular position of the plurality of photovoltaic solar panels (**10**), the method comprises the following steps: [0103] determining the initial angular position of the first set of panels (**11**); [0104] determining the initial angular position of the second set of panels (**12**); [0105] selecting a partial angular position of the plurality of photovoltaic solar panels (**10**), the partial angular position being comprised between at least one of the initial angular positions and the final angular position; [0106] moving the first set of panels (**11**) to the partial angular position depending on the initial angular position of the first set of panels (**11**); [0107] moving the second set of panels (**12**) to the partial angular position depending on the initial angular position of the second set of panels (**12**); and [0108] repeating the preceding steps until reaching the final angular position of the plurality of photovoltaic solar panels (**10**).
[0109] Clause 11. The control method according to clause 9 or 10, wherein the first set of panels (**11**) and the second set of panels (**12**) are moved simultaneously until reaching the final angular position.
[0110] Clause 12. The control method according to clause 9 or 10, wherein the first set of panels (**11**) and the second set of panels (**12**) are moved sequentially until reaching the final angular position, wherein first the photovoltaic solar panels (**10**) of one set of panels (**12**) are moved, and then the photovoltaic solar panels (**10**) of the other set of panels (**11**) are moved.
[0111] Clause 13. The control method according to the preceding clause, further comprising the following steps: [0112] determining a maximum torsion allowed by the main crossbar (**21**) of the support structure (**20**) on which the plurality of photovoltaic solar panels (**10**) is arranged, [0113] applying a first rotation to the first part (**23**) of the main crossbar (**21**) to move the first set of panels (**11**) without the first rotation exceeding the maximum torsion allowed by the main crossbar (**21**), [0114] applying a second rotation to the second part (**24**) of the main crossbar (**21**) to move the second set of panels (**11**) without the second rotation exceeding the maximum torsion allowed by the main crossbar (**21**), and [0115] repeating the application of the first rotation and the second rotation until reaching the final angular position of the plurality of photovoltaic solar panels (**10**).

Claims

1. A solar tracker comprising: a first set of photovoltaic solar panels; a second set of photovoltaic solar panels; a support structure that includes a main crossbar having a central longitudinal axis, the support structure also including a plurality of crossbars coupled to and oriented non-parallel to the main crossbar, the first and second sets of photovoltaic solar panels being coupled to the support structure, the main crossbar having a first part and a second part that is each independently rotatable with respect to the other; a first electric motor that is operatively coupled to the first part of the main crossbar to apply a rotation to the first part of the main crossbar to move the first set of photovoltaic solar panels to a final angular position; a second electric motor that is operatively coupled to the second part of the main crossbar to apply a rotation to the second part of the main crossbar to move the second set of photovoltaic solar panels to the final angular position; a first

position sensor configured to determine an initial angular position of the first set of photovoltaic panels; a first controller configured to control the first electric motor to rotate the first set of photovoltaic solar panels to the final angular position depending on the initial angular position determined by the first position sensor; a second position sensor configured to determine an initial angular position of the second set of photovoltaic solar panels; and a second controller configured to control the second electric motor to move the second set of photovoltaic solar panels to the final angular position depending on the initial angular position determined by the second position sensor.
2. The solar tracker according to claim 1, wherein at least some of the plurality of crossbars are oriented orthogonal to the main crossbar.
3. The solar tracker according to claim 1, further comprising a plurality of spaced apart posts on which the support structure is supported.
4. The solar tracker according to claim 1, wherein each of the first and second sets of photovoltaic solar panels is electrically coupled to a main power line to extract electric power generated by the first and second sets of photovoltaic solar panels, each of the first and second electric motors being electrically coupled to the main power line to receive a portion of the electric power from the power line.
5. The solar tracker according to claim 1, wherein the first controller is a master controller and the second controller is a slave controller, the first controller and the second controller being connected to one another through a data and power supply line.
6. The solar tracker according to claim 4, wherein the first controller is a master controller and the second controller is a slave controller, the first controller and the second controller being connected to one another through a data and power supply line that is configured to transmit electric power and communication data between the first and second controllers, the main power line being the data and power supply line.
7. The solar tracker according to claim 4, wherein the first controller is a master controller and the second controller is a slave controller, the first controller and the second controller being connected to one another through a data and power supply line, the first controller being electrically coupled to the main power line to receive electric power from the main power line, the second controller being electrically coupled to the first controller to receive electric power and communication data from the first controller through the data and power supply line.
8. The solar tracker according to claim 1, wherein each of the first and second position sensors is an accelerometer.
9. The solar tracker according to claim 8, wherein the first sensor is arranged in a first part of the support structure that supports the first set of photovoltaic solar panels, and the second sensor is arranged in a second part of the support structure that supports the second set of photovoltaic solar panels.
10. The solar tracker according to claim 9, wherein the first sensor is arranged in an enclosing casing of the first controller which is in turn arranged in the first part of the support structure, and the second sensor is arranged in an enclosing casing of the second controller which is in turn arranged in the second part of the support structure.
11. A control method of the solar tracker according claim 1, the control method comprising: selecting the final angular position; determining the initial angular position of the first set of photovoltaic solar panels; determining the initial angular position of the second set of photovoltaic solar panels; rotating the first set of photovoltaic solar panels to the final angular position depending on the initial angular position of the first set of photovoltaic solar panels; and rotating the second set of photovoltaic solar panels to the final angular position depending on the initial angular position of the second set of photovoltaic solar panels.
12. A control method of the solar tracker according to claim 1, the control method comprising: selecting the final angular position of the first and second sets of photovoltaic solar panels; determining the initial angular position of the first set of photovoltaic solar panels; determining the

initial angular position of the second set of photovoltaic solar panels; selecting a first partial angular position of the first and second sets of photovoltaic solar panels, the first partial angular position being comprised between at least one of the initial angular positions of the first and second sets of photovoltaic solar panels and the final angular position; rotating the first set of photovoltaic solar panels to the first partial angular position; rotating the second set of photovoltaic solar panels to the first partial angular position; selecting a second partial angular position of the first and second sets of photovoltaic solar panels, the second partial angular position being comprised between the first partial angular position and the final angular position; rotating the first set of photovoltaic solar panels to the second partial angular position; and rotating the second set of photovoltaic solar panels to the second partial angular position.

13. The control method according to claim 11, wherein the first set of photovoltaic solar panels and the second set of photovoltaic solar panels are moved simultaneously until reaching the final angular position.

14. The control method according to claim 11, wherein the first set of photovoltaic solar panels and the second set of photovoltaic solar panels are moved sequentially until reaching the final angular position, wherein the first set of photovoltaic solar panels is moved, and thereafter the second set of photovoltaic solar panels is moved.

15. A control method of the solar tracker according to claim 11, the control method comprising: determining a maximum torsion allowed by the main crossbar; applying a first rotation to the first part of the main crossbar to rotate the first set of photovoltaic solar panels without exceeding the maximum torsion allowed by the main crossbar; after the first rotation applying a second rotation to the second part of the main crossbar to rotate the second set of photovoltaic solar panels without exceeding the maximum torsion allowed by the main crossbar; and sequentially repeating the application of the first rotation and the second rotation until each of the first and second set of photovoltaic solar panels reach the final angular position.